

- 6 A cylindrical block of wood has a cross-sectional area  $A$  and weight  $W$ . It is completely immersed in water with its axis vertical. The block experiences pressures  $p_t$  and  $p_b$  at its top and bottom surfaces respectively.

Which of the following expressions is equal to the upthrust on the block?

A  $(p_b - p_t)A + W$

B  $(p_b - p_t)$

C  $(p_b - p_t)A$

D  $(p_b - p_t)A - W$

Ans: C

- 7 The engine of a boat supplies a constant power of 110 kW to propel the boat forward. The boat